



LANL Co-Design Summer School 2025

Model implosion dynamics at scale using FleCSI-HARD

Figure Credit: HARD CDSS24

Model various implosion dynamics and simulate the instabilities associated with these phenomena at scale on Venado:

- Base code is HARD, open source, developed by the Co-Design Summer School 2024 students
- Augment the code to model implosion: complex radiation-hydrodynamics scheme with different closures, set of solvers via FleCSolve, variable opacity, and verification + validation
- Optimize and profile the code at scale on the Venado supercomputer (Grace-Hopper architecture) to target large instances of inertial confinement fusion like problems

codesign.lanl.gov

The School:

- 10 weeks around mid-May to mid-August
- 6 students with multidisciplinary backgrounds (Not restricted to U.S. citizens)
- Working in a Co-Design manner on a common subject

Salary:

\$9-12k based on education and experience

Research areas:

- Computer Science (HPC)
- Rad-Hydro, Shock Physics
- Iterative/Multigrid Solvers



Send any questions to cdss@lanl.gov